



eWEngjry

A draft UAE-based Arab pangenome reference

Entry details

Title	Dr
First Name	Nasna
Last Name	Nassir
E-Mail	nasna.nassir@dubaihealth.ae
Mobile Number	+971586320112
Address of current residence	715, Azurite Tower, Jadaf, Dubai
City	Dubai
Country	United Arab Emirates
Designation	Assistant Professor
Organization	Mohammed Bin Rashid University of Medicine and Health Sciences, Dubai Health
Organization Address	Building 14, Dubai Healthcare City, P.O. Box 505055, Dubai, United Arab Emirates

Other Affiliations

Centre for Applied and Translational Genomics, Mohammed Bin Rashid University of Medicine and Health Sciences, Dubai Health

No Objection Letter

PDF	AARH Declaration - Signed... (367 KiB download)
-----	---

Authors Details

PDF	Author Details.pdf (94 KiB download)
-----	--

Curriculum Vitae

ZIP	Author Curriculum Vitae.... (22.6 MiB download)
-----	---

Research Paper

PDF	A draft UAE-based Arab pa... (2.2 MiB download)
-----	---

Online Publication Link <https://doi.org/10.1038/s41467-025-61645-w>

Has the submitted publication received any awards? No

Layperson's Summary

The Arab Pangenome Reference study addresses a major gap in global genomics: the underrepresentation of Arab populations in existing human genome references. Most current genome references are built from limited population backgrounds and may miss DNA sequences, variants, and structural changes that are more common or specific to Arab individuals. This limitation can affect genetic discovery, disease research, and the accuracy of clinical genome interpretation in the region.

In this study, we developed a draft UAE-based Arab Pangenome Reference using high-quality genome data from 53 healthy individuals of diverse Arab backgrounds residing in the UAE. By combining advanced long-read and chromosome-conformation sequencing technologies, we generated high-quality phased genome assemblies and constructed nuclear and mitochondrial pangenome references. The study identified substantial previously uncharacterized genetic diversity, including novel sequences, population-specific small variants, structural variants, gene duplications, and mitochondrial sequences that are absent from existing reference genomes and public datasets.

This work provides a foundational resource for precision medicine in Arab populations. It can improve the interpretation of genetic variants, support research into rare and complex diseases, and reduce bias in genomic analyses caused by reliance on non-representative reference genomes. The reference and associated analytical framework also provide a platform for future regional and global genomic

studies, helping ensure that Arab populations are more accurately represented in the next generation of human genome references.

Institute/Facility where the Research was primarily Conducted	Centre for Applied and Translational Genomics, Mohammed Bin Rashid University of Medicine and Health Sciences, Dubai Health
---	---

Institute address and contact person

Centre for Applied and Translational Genomics
Mohammed Bin Rashid University of Medicine and Health Sciences
Building 14, Dubai Healthcare City
P.O. Box 505055, Dubai, United Arab Emirates

Contact person:
Dr. Mohammed Uddin
Director, Centre for Applied and Translational Genomics
Email: Mohammed.Uddin@dubaihealth.ae
Phone: 971-582478565

Scientific Robustness and Originality - 30%

/ 5.00

5 being the highest and 1 as the lowest. * The study design and approach are clear and appropriate. * Findings are robust and reproducible. * Research is innovative and original.

Impact - 55%

/ 5.00

5 being the highest and 1 as the lowest. * The research findings are highly relevant and address prevailing healthcare issues. * The results contribute new knowledge to the field. * It has the potential to make a significant impact on the field. * Findings contribute to policy changes or future practice. * Findings may benefit patient care or public health. * The paper is well-cited.

Future research directions - 15%

/ 5.00

5 being the highest and 1 as the lowest. * The research outlines potential avenues for further research.

Log in to hmaward.awardsplatform.com to see complete entry attachments.

PDF
Description Official Dubai Health/MBRU news release announcing the publication of the first Arab Pangenome Reference in Nature Communications, highlighting its relevance to precision medicine, the UAE National Genome Strategy, CATG infrastructure, and regional genomic healthcare.
Dubai Health sc... 1.1 MiB

PDF
Description Technical press release from PacBio highlighting the role of HiFi long-read sequencing in the first Arab human pangenome study, including the discovery of novel sequences, population-specific variants, structural variants, and its relevance to precision medicine.
PacBio HiFi Seq... 275 KiB

PDF
Description External scientific/media coverage highlighting the significance of the first UAE-based Arab Pangenome Reference, including its contribution to Arab genomic representation, precision medicine, variant discovery, and future healthcare applications.
The Pathologist ...506 KiB


Attachment name External scientific/medi...
https://www.dubaieye1...


Attachment name External scientific/medi...
https://english.mathrub...